

ARPAN POUDEL

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OBJECTIVE

Ph.D. student with expertise in machine learning, computer vision, federated learning, and multimodal model development. Experienced in designing, implementing, and evaluating deep learning models for healthcare applications.

EDUCATION

Ph.D. - Computer Science University of Arkansas, Fayetteville, AR	Dec 2027 4.0 GPA
Master of Science (M.S.) - Computer Science University of Arkansas, Fayetteville, AR	Dec 2024 4.0 GPA
Bachelor of Engineering (B.E.) - Computer Engineering Tribhuvan University, Kathmandu, Nepal	Dec 2018 74 %

SKILLS

Programming Languages: Python, C++, Java, MATLAB

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, NumPy, Pandas, Scikit-learn, OpenCV, Seaborn

Machine Learning & AI: Deep Learning, Computer Vision, Large Language Models (LLMs), Generative AI, Feature Engineering, Model Optimization, Image Processing, EDA

Systems: Federated Learning, Distributed Training, Model Deployment, HPC, CUDA

Databases: SQL, MongoDB, PySpark

Tools: Git, Docker, Jupyter Notebook, VS Code, Excel, Microsoft Office, Heroku

EXPERIENCE

University of Arkansas, Fayetteville, AR: Research and Teaching Assistant	Jan 2022 – Present
- Developed an efficient routing framework for selectively reusing pretrained experts in federated learning settings. - Designed and implemented a federated learning algorithm to utilize adapters trained across distributed clients, enabling a privacy-preserving vision-language model for downstream tasks including radiology report generation and zero-shot classification on decentralized chest X-ray datasets. - Developed deep learning models for MRI super-resolution to support the study of brain microstructure in Alzheimer's disease. - Led a lensless imaging project, addressing inverse problems using physics-informed, attention-based neural network. - Assisted in teaching undergraduate courses including Algorithms, Databases, Operating Systems, and Software Engineering.	
Government of Nepal, Jhapa, Nepal: IT Engineer	Nov 2019 – Dec 2021
Designed and led the development of a centralized application to streamline service management in an e-governance system.	

PROJECTS

Federated Vision-Language Models for Chest X-ray Analysis	Jan 2025 – Present
Fine-tuned multimodal models on distributed MIMIC-CXR-JPG data for zero-shot classification and report generation in privacy-preserving settings.	
- Implemented a federated training pipeline using Hugging Face Transformers and Flower for privacy-preserving fine-tuning of vision-language models. - Adapted LoRA adapters trained across clients to generate reports from Chest X-ray images. - Evaluated cross-client performance using decentralized metrics logging with Weights & Biases.	
Unsupervised Magnetic Resonance Imaging Super-resolution	Jan 2024 - Dec 2024
Proposed and developed a novel unsupervised deep learning network for Magnetic Resonance Imaging (MRI) capable	

of generating high-resolution images from low-resolution images which is 5 times faster than current state-of-the-art methods.

- Trained generative model to learn the distribution of medical images to sample high-resolution MRI images reducing the MRI scan time by 30% in multi-node multi-GPUs.
- Prepared and preprocessed a medical dataset to train a custom generative model.
- Proposed a novel sampling algorithm to enforce data consistency using the Wavelet Transform.

Solving inverse problems in lensless image reconstruction with generative models Aug 2023 - Jan 2024

Created a novel sampling algorithm for controllable image generation for lensless cameras.

- Trained score-based generative model to learn the distribution of natural images as a prior achieved 1.2 times increase in image reconstruction accuracy.
- Implemented an unrolling of the Alternating Direction Method of Multipliers (ADMM) for data consistency.

PUBLICATIONS

- **A. Poudel**, M. Acharya, S. Panta, B. Bhusal, Q. Li “*FedRoute: Selective Federated Learning via Routing over Client Experts*,” Preprint.
- S. Panta, **A. Poudel**, Q. Li “*Beyond Approximation Error: A Noise-Aware Polynomial Design for Encrypted Neural Inference*,” Preprint.
- **A. Poudel**, M. Shrestha, U. Nakarmi, N. Wang, “*SRMRI: A Diffusion-Based Super-Resolution Framework and Open Dataset for Blind MRI Super-Resolution*,” in Medical Imaging with Deep Learning (MIDL), 2025.
- H. Pradhan, **A. Poudel**, D. Shrestha, A. Rogers, M. Stewart, A. Jereb, J. Harper, M. Li, W. Zhang, J. Chen, Y. Wang, “*Concentration-Dependent Responses of C. reinhardtii to Silver Ions: Hormetic Response in Growth and Reduction of Motility*,” The European Physical Journal E, 2025.
- **A. Poudel**, U. Nakarmi, “*DeepLIR: Attention-Based Approach for Mask-Based Lensless Image Reconstruction*,” in Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision W, 2024.
- E. Kabir, **A. Poudel**, Z. Aklah, M. Huang, D. Andrews, “*A Runtime Programmable Accelerator for Convolutional and Multilayer Perceptron Neural Networks on FPGA*,” in Applied Reconfigurable Computing, 2022.
- A. Dhakal, **A. Poudel**, S. Pandey, S. Gaire, H. P. Baral, “*Exploring Deep Learning in Semantic Question Matching*,” in IEEE International Conference on Computing, Communication and Security (ICCCS), 2018.

VOLUNTEER

- Reviewed submitted papers for conferences, including SSIAI 2024, ICECER 2024, and ICECCME 2024.
- Led and managed the Nepalese Student Organization at the University of Arkansas.

AWARDS

- Won first prize of \$2,500 in the Healthcare Innovation Sprint 2024, organized by the Conductor, Arkansas.
- Reginald R. “Barney” & Jameson A. Baxter Graduate Fellowship 2023.
- First Runner-Up in Genese CodeCamp 2018 Coding Competition.

MEMBERSHIP

- IEEE
- IEEE Computer Society